A Comprehensive Evaluation of Time-Series Forecasting Methods for Proactive Handover

Hyeryn Park
Department of Artificial Intelligence
Chung-Ang University
Seoul, Korea
hyeryn2000@cau.ac.kr

Eungyeop Kim
Department of Artificial Intelligence
Korea University
Seoul, Korea
key2317@korea.ac.kr

Changhee Lee
Department of Artificial Intelligence
Korea University
Seoul, Korea
changheelee@korea.ac.kr

2024 15th International Conference on Information and Communication Technology Convergence (ICTC) | 979-8-3503-6463-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICTC62082.2024.10827118

***Abstract*—With the advancement of 5G and beyond, mobility management has become increasingly complex, driving the need for advanced proactive handover (HO) strategies. Traditionally, HO algorithms have been reactive, but recent advancements in AI/ML offer proactive solutions through time-series forecasting. In this paper, we present a comprehensive comparison of state-of-the-art time-series forecasting methods essential for proactive HO within the 3rd Generation Partnership Project simulation environment. Instead of focusing solely on overall prediction accuracy, we assess performance across various HO-related metrics to gain a thorough understanding. Additionally, we introduce a novel masking mechanism that prioritizes potential target cells and specific time points critical for HO decisions, enhancing the predictive power of time-series methods. This approach leads to improved HO performance and underscores the need for further exploration in this area.**

***Index Terms*—Mobility management, RSRP measurement prediction, proactive handover, time-series forecasting, deep learning**

I. Introduction

Mobility management in cellular networks has become increasingly critical with the core advancements in wireless communication, aiming to provide uninterrupted service even as users traverse across different cells. The advent of 5G and beyond has introduced new challenges, particularly those stemming from denser network deployments, the utilization of higher frequency bands, and the stringent requirements of ultra-reliable low-latency communications (URLLC) [1].

Traditionally, mobility management has relied on reactive approaches, responding to changes in received signal strength or when a certain failure scenario occurs. However, the growing complexity of modern networks and stringent user requirements have necessitated a shift towards more sophisticated, proactive methods. This transition aims to minimize service interruptions, reduce signaling overhead, and optimize overall network performance. To address these challenges, the 3rd Generation Partnership Project (3GPP) has initiated studies [2] into the potential enhancements offered by proactive handover strategies in mobility management, leveraging the power of Artificial Intelligence and Machine Learning (AI/ML) [3, 4, 5, 6, 7]. These approaches aim to predict network conditions and user movements, enabling anticipatory decision-making in mobility management. By utilizing AI/ML for mobility prediction, proactive HO mechanisms seek to optimize HO timing and target cell selection, potentially preventing handover failures (HOF) and ping-pong [8] effects before they occur.

Despite these advancements, current proactive HO approaches have limitations. They often focus on improving time-series forecasting performance across all serving cells, relying on restricted metrics for evaluation. Moreover, these approaches typically employ a single AI/ML model for both *short-term* (Time-To-Trigger, TTT) and *long-term* (Minimum Time of Stay, MTS) predictions, potentially overlooking the unique characteristics of each time horizon (e.g., differences in short-term and long-term temporal dynamics).

In this paper, we aim to explore more practical applications of AI/ML for proactive HO from the following perspectives:

- **Model Comparison:** We conduct a comprehensive comparison of various state-of-the-art time-series models for forecasting, ranging from established RNN-based models to newer transformer and segment-based architectures.
- **Targeted Prediction:** Instead of evaluating performance across all cells, we prioritize accuracy in predicting specifically for candidate cells, those identified as potential HO targets.
- **Evaluation Metrics:** In addition to traditional mean absolute error (MAE) and root mean squared error (RMSE) metrics, we assess top-1 margin accuracy. This complements existing metrics by focusing on the successful identification of the optimal target cell, which aligns better with the HO objective.

Contribution. We delve into the practical application of cutting-edge time-series forecasting methods within the context of proactive HO. Our study provides a comprehensive analysis that addresses limitations observed in prior research. Furthermore, we introduce an innovative masking mechanism that enhances top-1 margin accuracy by selectively predicting received signals only from cells with signal strength exceeding a defined threshold. Furthermore, we refine our approach by targeting specific time points – i.e., TTT and MTS – for forecasting, rather than predicting across all time points up to MTS. This targeted strategy allows us to develop models specifically optimized for predicting these critical HO moments, ultimately leading to improved top-1 margin accu-

Authorized licensed use limited to: International Institute of Information Technology-Raipur. Downloaded on November 30,2025 at 13:55:09 UTC from IEEE Xplore. Restrictions apply.

racy, which is crucial for enhancing overall HO performance. Through extensive experimentation, we thoroughly compare state-of-the-art time-series forecasting methods using multiple HO-related metrics, offering insights into how to effectively integrate AI/ML techniques into proactive HO.

II. Related works

A. Mobility Management and Proactive HO

Mobility management is a fundamental feature in cellular networks, aiming to maintain service continuity during user mobility. It focuses on minimizing cell drops, radio link failures (RLFs), unnecessary HOs, HOFs, and ping-pong effects. Traditionally, mobility robustness optimization has been used to automate the selection of network configuration parameters. As networks have evolved, particularly with the advent of 5G and beyond, the complexity of mobile management has increased. This evolution has led to the development of two primary approaches in mobile management systems from user equipment (UE) and network (NW) perspectives. UE-side models leverage more frequent and accurate measurement data, while NW-side models rely on centralized control based on less frequent measurement feedback from UEs.

Among the various aspects of mobility management, HO plays a crucial role in ensuring seamless connectivity as UEs move across different cells. Proactive HO algorithms, in particular, have recently gained attention for their ability to anticipate optimal HO timing and target cell selection by anticipating the future, typically leveraging AI/ML techniques. This proactive approach can significantly reduce HOFs and ping-pong effects, ultimately improving user-perceived throughput.

B. Time-Series Forecasting

Time-series forecasting, which predicts future measurements based on historical data points ordered in time, has become a critical area of research in machine learning, spanning various domains including finance, weather prediction, and telecommunications. In the context of proactive HO, time-series forecasting plays a pivotal role in predicting network parameters such as RSRP measurements to determine the optimal timing and target for HO.

The authors in [9] have laid important milestones in applying time-series forecasting to proactive HO. However, these studies often fall short of utilizing the latest advancements in deep learning-based time-series models. Our goal is to investigate the real-world applicability of these advanced baselines and provide a comprehensive comparison and analysis of not only forecasting performances but also the performance differences based on the proposed methods.

This study evaluates a range of AI/ML models designed for multivariate time-series forecasting in the context of proactive HO algorithms. We consider recurrent neural networks, specifically Long Short-Term Memory (LSTM) [10], Gated Recurrent Units (GRU) [11], and SegRNN [12], which are known for their ability to capture long-term dependencies in sequential data. In the realm of convolutional networks, we examine Temporal Convolution Networks (TCN) [13] as well as PatchMixer [14], a novel CNN-based model that introduces a permutation-variant convolutional structure to preserve temporal information. Additionally, we assess transformer-based architectures, including Informer [15] and Autoformer [16], which have shown promising results in handling long sequence predictions. Lastly, we include N-Linear [17] and D-Linear [17], a single-layer model that has demonstrated competitive performance despite its linear simplicity.

In alignment with the ongoing exploration of AI/ML Management for 5G Systems by the 3GPP, our research focuses on applying and evaluating cutting-edge time-series forecasting methods within a 3GPP-compliant simulation environment. We particularly emphasize multivariate time-series data to capture the complex dynamics inherent to mobility management. Through these standardized experiments, we assess the efficacy of these advanced forecasting models and investigate their practical impact when considering different scenarios, offering insights for optimizing mobility management as future research.

III. Preliminaries

A. RSRP measurement prediction as time-series forecasting

Proactive HO tackles the trade-off between HO failures (HOFs) and ping-pong effects by predicting future network conditions based on the forecasting of Reference Signal Received Power (RSRP) measurements, which are periodically observed by user equipment (UE). Predicting future RSRP measurements from historical observations can be formulated as a discrete-time multivariate time-series forecasting problem. Suppose we are given a time-series sample represented as $\mathbf{X}_{1:N,1:T} \in \mathbb{R}^{N\times T}$ (we will occasionally omit the dependency on N and T and denote $\mathbf{X}$ to represent the full sequence when clear in the context), which is a sequence of T RSRP measurements (i.e., RSRP measurements observed over T time steps) from N cells. Formally, we can define a length-W_{out} measurement prediction problem at each time step t given a length-W_{in} observed history of measurements as input:

$$\hat{\mathbf{X}}_{1:N,t+1:t+W_{\text{out}}+1} = f_\theta(\mathbf{X}_{1:N,t-W_{\text{in}}+1:t}), \tag{1}$$

where W_{in} and W_{out} are look-back and look-forward windows, respectively. Here, $f_\theta : \mathbb{R}^{N\times W_{\text{in}}} \rightarrow \mathbb{R}^{N\times W_{\text{out}}}$ is a function, parameterized θ, that maps a multivariate time-series input to a multivariate time-series output. In multivariate time-series forecasting, we aim to achieve f_θ that minimizes a loss function $\mathcal{L}_X(\mathbf{X}_{N,t+1:t+W_{\text{out}}+1}, \hat{\mathbf{X}}_{N,t+1:t+W_{\text{out}}+1})$, such as MAE or RMSE, between the predicted and ground-truth RSRP measurements.

B. Challenges

While the multivariate time-series forecasting problem in (1) focuses on minimizing the prediction error for the RSRP measurements of all cells across the entire look-forward window (W_{out}), there are three key perspectives that require careful consideration for the practical application of AI/ML in proactive HO.

Authorized licensed use limited to: International Institute of Information Technology-Raipur. Downloaded on November 30,2025 at 13:55:09 UTC from IEEE Xplore. Restrictions apply.

- **Targeted time point prediction.** The primary concern in proactive HO is the accurate prediction at specific time points: TTT and MTS, representing short-term and long-term time horizons, respectively. These points are critical. Short-term forecasting at TTT predicts immediate signal changes, enabling proactive handovers and reducing handover failures. Long-term forecasting at MTS prevents ping-pong effects, ensuring sustained connections to optimal cells. Therefore, we propose prioritizing effective learning at these specific target time points, rather than evaluating across all time points. This approach aims to develop models specifically optimized for predicting TTT and MTS, thereby enhancing their overall performance in the context of proactive HO.
- **Target cell candidates.** HO is exclusively executed to a selected target cell within the designated target cell candidates. This implies that it might be unnecessary to track and predict RSRP measurements for all available cells. Instead, prioritizing accurate predictions for the top-ranked cells (potential HO targets), rather than treating all cells equally, can be crucial. This targeted approach allows the forecasting models to concentrate on the most relevant cells, reducing the impact of irrelevant signals that could otherwise affect prediction performance.
- **Top-1 prediction accuracy.** Traditional proactive HO frameworks typically rely on metrics like MAE and RMSE evaluated across all cells. However, this approach does not fully align with the core objective of proactive HO, which is to accurately pinpoint candidate cells with sufficient signal strength to ensure successful HO. Therefore, a more suitable evaluation involves assessing whether the predicted strongest cell offers a signal strong enough when compared to the actual (ground-truth) strongest cell in the ground truth.

IV. Methods

In this section, we present two adaptations of the multivariate time-series forecasting problem outlined in (1). The first modification focuses on prioritizing potentially significant cells, while the second modification targets two specific time points of interest: TTT and MTS.

A. Prioritizing Potentially Significant Cells

In proactive HO algorithms, the key is to pinpoint the best handover target by focusing on cells that show promise for the highest future RSRP measurements. It is unnecessary to track and predict RSRP measurements for all available cells. We suggest a masking mechanism to guide the forecasting process toward these potentially important cells. We prioritize cells with strong RSRP measurements, as they are the most promising candidates to offer the best connection in the future.

Suppose $S_t = \mathbf{X}_{n^*,t}$ represents the RSRP measurements of the serving cell at a given time step t where n^* is the index for the serving cell. Our masking mechanism will then mask any cells whose RSRP measurements at that time are lower than the serving cell's RSRP measurements by a certain pre-specified threshold, denoted by α (e.g., 15 dB).

Formally, let $\mathbf{M} \in \{0,1\}^{N \times W_{\text{out}}}$ be the mask vector. We define our mask vector as the following to filter out the less promising candidate cells:

$$\mathbf{M}_{n,:} = \begin{cases} \mathbf{1} & \text{if } \mathbf{X}_{n,t} + \alpha > S_t \\ \mathbf{0} & \text{otherwise} \end{cases} \quad \text{for } n \in \{1, \ldots, N\} \quad (2)$$

By applying the mask vector to the loss function as $\mathcal{L}_X(\mathbf{X} \odot \mathbf{M}, \hat{\mathbf{X}} \odot \mathbf{M})$, we direct the time-series forecasting methods to focus their predictions specifically on those potential HO target cell candidates. This effectively narrows down the model's focus and optimizes its performance. It is worth highlighting that evaluation metrics such as top-1 accuracy and RSRP measurements within a 5dB range of the highest observed values will lead to a more targeted and effective evaluation of HO scenarios.

B. Prioritizing Targeted Time Points

To trigger a HO, certain conditions, like a neighboring target cell being an offest better than the serving cell (Event A3), need to be fulfilled at specific time points (i.e., TTT and MTS). Hence, AI/ML model enhancements for proactive HO should prioritize RSRP measurement predictions at these short-term (TTT) and long-term (MTS) time points. To this goal, we prioritize these specific target time points, rather than across all time points, by introducing the following masking mechanism:

$$\mathbf{M}_{:,t} = \begin{cases} 1 & \text{if } \quad t \in \{t_1, t_2\} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

where t_1 and t_2 indicate the time points at TTT (160 ms) and MTS (1000 ms), respectively. Similarly, we apply the mask vector to the loss function as $\mathcal{L}_X(\mathbf{X} \odot \mathbf{M}, \hat{\mathbf{X}} \odot \mathbf{M})$. By doing so, we direct the time-series forecasting methods to focus on those time points, pivotal for HO decision-making.

V. Experiments

A. Datasets

We evaluate various time-series forecasting methods through simulations, complying with the scenario for large area system simulations in [2], i.e., 3GPP two-tiers macro cell scenario with ISD of 500m (19 BSs, 57 Cells), using OPNET LTE model. The UE speed of 60 km/h and the random waypoint mobility model is used. The measurement period of 40 ms and L3 filter coefficient of 1 are used. We partition the UE trajectory into non-overlapping sequences, allocating them into training, validation, and testing sets in a 64:16:20 ratio. Each sequence is further partitioned into input and output subsequences, each of length 32, using a stride of 1.

B. Multivariate Time-series Forecasting Methods

Our evaluation includes a range of state-of-the-art multivariate time-series methods as baselines, encompassing:(i) RNN-based methods such as **LSTM** [10], **GRU** [11], and **SegRNN** [12], (ii) CNN-based methods such as **TCN** [13]

Authorized licensed use limited to: International Institute of Information Technology-Raipur. Downloaded on November 30,2025 at 13:55:09 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PERFORMANCE COMPARISONS FOR CELLS WITHIN 5DB BASED ON THE SERVING CELL.

	TTT (160ms)			MTS (1000ms)		
Models	Acc	MAE	RMSE	Acc	MAE	RMSE
LSTM	54.851	5.153	5.999	54.654	6.388	6.388
GRU	52.304	4.407	5.431	52.126	4.551	6.078
SegRNN	94.576	1.261	1.978	56.673	**1.889**	**1.891**
TCN	94.469	1.118	1.359	**57.385**	3.340	4.231
PatchMixer	**94.920**	**0.915**	**0.908**	56.541	2.081	2.084
Informer	81.286	3.050	3.582	53.973	3.605	3.971
Autoformer	85.279	2.689	3.003	53.545	3.586	3.591
DLinear	94.894	0.918	0.919	52.519	4.455	5.790
NLinear	94.898	1.108	1.443	57.058	2.081	2.139

TABLE II
PERFORMANCE COMPARISONS FOR CELLS WITHIN 5DB MARGIN BASED ON THE SERVING CELL (APPLYING SPECIFIC DB MASKING FOR EACH MODEL).

		TTT (160ms)			MTS (1000ms)		
Models	dB	Acc	MAE	RMSE	Acc	MAE	RMSE
LSTM	45	54.998	4.645	5.167	55.545	4.071	4.373
GRU	45	52.036	1.714	2.342	52.026	3.198	4.788
SegRNN	45	94.705	1.269	1.337	56.715	1.650	1.682
TCN	15	94.768	1.046	1.236	**59.808**	2.966	3.612
PatchMixer	35	94.846	0.907	1.045	56.023	1.879	1.886
Informer	35	81.220	1.884	3.198	53.252	3.618	4.288
Autoformer	35	83.973	1.203	1.528	55.658	**1.532**	**1.920**
DLinear	5	**95.135**	**0.876**	**0.898**	59.653	3.235	4.036
NLinear	5	94.871	1.131	1.235	57.789	1.981	1.991

and **PatchMixer** [14], (iii) transformer-based models such as **Informer** [15] and **Autoformer** [16], and (iv) recent long-term forecasting linear models such as **DLinear** [17] and **NLinear** [17]. Notably, the average inference time of all the aforementioned methods is below 10ms[1], which we believe acceptable for HO decision-making.

C. Evaluation Metrics

In our experiments, we assess the performance of RSRP measurement prediction using multiple evaluation metrics. These metrics focus on two key aspects: **(Prediction Errors)** We quantify how closely the predicted RSRP measurements align with the actual measurements obtained from *genie-aided* measurements, which possess perfect knowledge of future RSRP values. **(Accuracy)** We evaluate how well the model can identify the cells with the strongest RSRP measurements based on the genie-aided measurements. This highlights the model's ability to pinpoint the most promising cells for HO decisions. We provide their mathematical definitions below:

- **(Prediction Errors)** We measure the average L1-RSRP difference between our predictions and the actual ground-truth values based on the top-1 cell identified by the genie-aided measurements. Two prediction error metrics used in this work include L1-RSRP MAE, which is calculated as [18]:

$$\frac{1}{m}\sum_{i=1}^{m}|\mathbf{X}^i_{n^*,t}-\hat{\mathbf{X}}^i_{n^*,t}|$$

and L1-RSRP RMSE, which is computed as:

$$\sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(\mathbf{X}^i_{n^*,t}-\hat{\mathbf{X}}^i_{n^*,t}\right)^2},$$

where $n^* = \arg\max_n \mathbf{X}_{n,t}$ be the top-1 cell based on the genie-aided RSRP measurements and m indicates the number of test samples.

- **(Accuracy)** We introduce the top-1 margin accuracy, which evaluates whether the ground-truth RSRP measurements of the cell predicted to have the highest RSRP is within a 5dB margin when compared to the ground-truth RSRP measurements of the top-1 cell identified by the genie-aided measurements. Formally, we can compute this metric as

$$\frac{1}{m}\sum_{i=1}^{m}\mathbb{1}\left(|\mathbf{X}^i_{n^*,t}-\mathbf{X}^i_{\hat{n},t}|<5\right)$$

where $n^* = \arg\max_n \mathbf{X}_{n,t}$ and $\hat{n} = \arg\max_n \hat{\mathbf{X}}_{n,t}$ be the top-1 cell based on the genie-aided RSRP measurements and based on the RSRP predictions, respectively.

[1]The inference time was measured by averaging 100 samples on a machine with an INTEL XEON Gold 6240R and an NVIDIA RTX A6000.

TABLE III
PERFORMANCE COMPARISONS FOR CELLS WITHIN 5DB MARGIN BASED ON THE SERVING CELL (APPLYING SPECIFIC DB MASKING FOR EACH MODEL AND TARGETING SPECIFIC TTT AND MTS).

		TTT (160ms)			MTS (1000ms)		
Models	dB	Acc	MAE	RMSE	Acc	MAE	RMSE
LSTM	45	55.312	4.459	5.269	54.225	5.846	6.126
GRU	45	50.203	2.958	3.483	50.998	6.073	6.037
SegRNN	45	94.863	1.195	1.298	**56.580**	1.587	**1.653**
TCN	15	**95.097**	**0.961**	1.054	59.691	2.764	3.407
PatchMixer	35	94.488	1.169	1.206	56.010	1.958	1.958
Informer	35	81.727	2.222	3.472	54.224	3.568	3.693
Autoformer	35	86.545	2.330	2.762	52.656	**1.502**	1.965
DLinear	5	95.073	0.991	**1.030**	51.737	4.688	5.988
NLinear	5	94.937	0.974	1.031	55.941	1.672	1.705

D. Experiment Results

We showcase the performance of the time-series forecasting methods in Tables I, II, and III. We highlight the best-performing method in bold and we underline the top performance for each method across the three tables.

There are a couple of things to be highlighted from these tables: First, in contrast to the baseline performance shown in Table I, applying our proposed masking strategy, prioritizing the potential target cells as in (2), results in significant improvements across most models, particularly in short-term predictions as shown in Table II. Second, contrasting Table II

Authorized licensed use limited to: International Institute of Information Technology-Raipur. Downloaded on November 30,2025 at 13:55:09 UTC from IEEE Xplore. Restrictions apply.

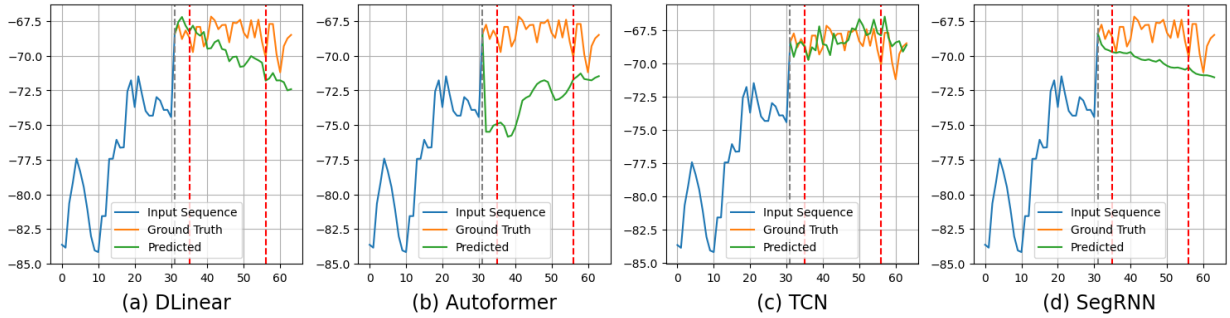

Fig. 1. Default prediction results are shown, utilizing various baseline models. The blue, orange, and green lines denote the input, ground-truth future, and prediction values, respectively. The gray dashed line is the time of prediction and the red dashed line indicates TTT and MTS time points.

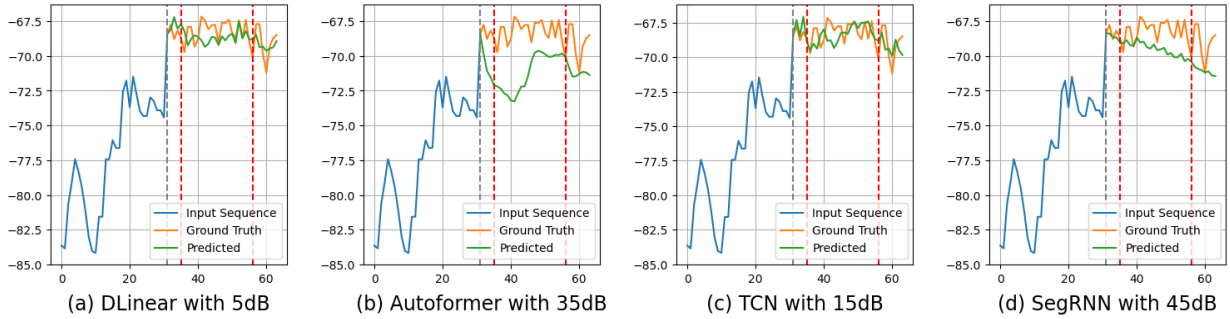

Fig. 2. Prediction results using various baseline models, prioritizing strongest measurements within a specific dB margin based on the serving cell.

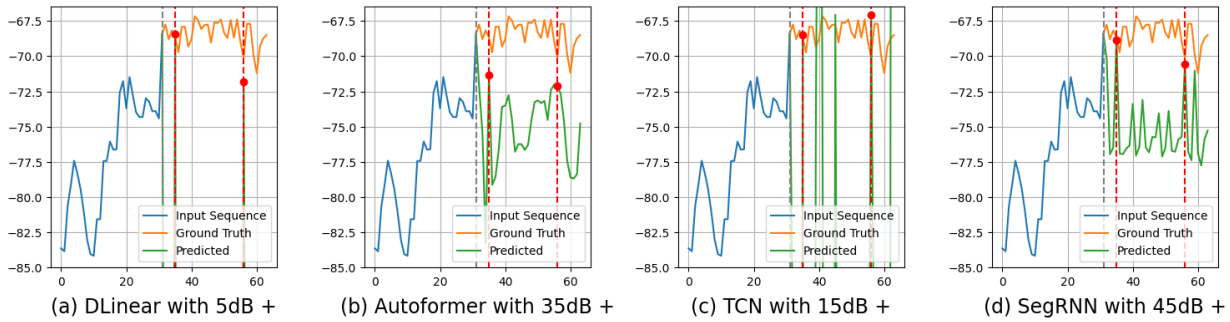

Fig. 3. Prediction results using various baseline models, prioritizing both the strongest measurements within a specific dB margin and the target points i.e., TTT and MTS (indicated by $+$ sign).

and Table III reveals that focusing on specific time points lessens the performance drop observed between the short-term to long-term predictions. This targeted approach consistently helps to minimize the overall performance drop, albeit to different extents depending on the model. We delve deeper into these implications in the following.

The impact of pre-specified thresholds. Table II shows the importance of selecting an appropriate threshold in (2), emphasizing the need to consider model-specific characteristics for masking strategy. The threshold α is selected from a set of candidate $\{5, 15, \ldots 95\}$ using a grid search. Linear-based methods tend to yield better results when focusing on small dB margins. We speculate that their simple structure allows for generalization with less data, making them more robust even when learning based on a small portion of target cells. In contrast, both the transformer-based and CNN-based benefit from increasing exposure with larger dB margins. From this, we suggest that excessive masking is not always advantageous and highlights the importance of considering model-specific characteristics when deciding on the masking strategy. Specifically, Figure 2 shows that setting appropriate margins to focus on top RSRP cells elevates model predictions compared to the baseline in Figure 1. This demonstrates that tailoring margins based on the model's characteristics enhance predictive accuracy.

The impact of targeted prediction. Table III shows the evaluation results for short-term (TTT) and long-term (MTS) forecasting using the same thresholds as in Table II. While most models show performance improvements, particularly in long-term forecasting, the performance of LSTM and GRU has decreased. We speculate that this degradation might be attributed to their inherent recurrent structure, which depends on maintaining a memory state across sequences. That is, when forced to focus on specific time points, these models may

Authorized licensed use limited to: International Institute of Information Technology-Raipur. Downloaded on November 30,2025 at 13:55:09 UTC from IEEE Xplore. Restrictions apply.

struggle to effectively propagate relevant information, leading to a loss of crucial contextual data over far future sequences. In contrast, SegRNN and TCN demonstrated improved performance in both short-term and long-term predictions. More specifically, SegRNN likely benefits from its segment-wise prediction approach, which, combined with masking, allows for more focused learning on each segment of the time-series data, thus preserving important contextual information.

These results emphasize the significance of employing model-specific strategies, such as adjusting dB masking thresholds or focusing on specific time points like TTT and MTS. Figure 3 further illustrates how targeted masking can lead to improved accuracy at these critical moments.

VI. Conclusion

This study delivers a comprehensive comparative analysis of cutting-edge time-series methods applied to AI/ML-enhanced proactive HO strategies in cellular networks. Our findings underscore the importance of targeting predictions toward potential target cell candidates and specific time points, overcoming limitations associated with conventional multivariate time-series forecasting approaches. Additionally, we observe that model-specific differences may arise from different masking strategies, highlighting a promising avenue for future research to explore and refine model-dependent prioritization on target RSRP measurements to optimize proactive HO.

Acknowledgments

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No. 2021-0-00972, Development of Intelligent Wireless Access Technologies)

References

[1] H.-S. Park, Y. Lee, T.-J. Kim, B.-C. Kim, and J.-Y. Lee, "Handover mechanism in nr for ultra-reliable low-latency communications," *IEEE Network*, vol. 32, no. 2, pp. 41–47, 2018.

[2] "E-utra; mobility enhancements in heterogeneous networks (release 11)," 2012, standard 3GPP TR 36.839.

[3] S. M. A. Zaidi, H. Farooq, A. Rizwan, A. Abu-Dayya, and A. Imran, "A framework to address mobility management challenges in emerging networks," *IEEE Wireless Communications*, vol. 30, no. 4, pp. 90–97, 2023.

[4] H. Rydén, H. Farhadi, A. Palaios, L. Hévizi, D. Sandberg, and T. Kvernvik, "Next generation mobile networks' enablers: Machine learning-assisted mobility, traffic, and radio channel prediction," *IEEE Communications Magazine*, vol. 61, no. 10, pp. 94–98, 2023.

[5] C. Lee, H. Cho, S. Song, and J.-M. Chung, "Prediction-based conditional handover for 5g mm-wave networks: A deep-learning approach," *IEEE Vehicular Technology Magazine*, vol. 15, no. 1, pp. 54–62, 2020.

[6] J. P. Lima, Á. A. de Medeiros, E. P. de Aguiar, E. F. Silva, V. A. de Sousa, M. L. Nunes, and A. L. Reis, "Deep learning-based handover prediction for 5g and beyond networks," in *ICC 2023-IEEE International Conference on Communications*. IEEE, 2023, pp. 3468–3473.

[7] A. Masri, T. Veijalainen, H. Martikainen, S. Mwanje, J. Ali-Tolppa, and M. Kajó, "Machine-learning-based predictive handover," in *2021 IFIP/IEEE International Symposium on Integrated Network Management (IM)*. IEEE, 2021, pp. 648–652.

[8] "Study on enhancement for data collection for nr and endc (release 17)," 2022, standard 3GPP TR 37.817.

[9] H.-S. Park, H. Kim, C. Lee, and H. Lee, "Mobility management paradigm shift: From reactive to proactive handover using ai/ml," *IEEE Network*, vol. 38, no. 2, pp. 18–25, 2024.

[10] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[11] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," in *Proceedings of the Neural Information Processing Systems Deep Learning and Representation Learning Workshop (NeurIPS Workshop 2014)*, 2014.

[12] S. Lin, W. Lin, W. Wu, F. Zhao, R. Mo, and H. Zhang, "Segrnn: Segment recurrent neural network for long-term time series forecasting," *arXiv preprint arXiv:2308.11200*, 2023.

[13] Y. Lin, I. Koprinska, and M. Rana, "Temporal convolutional attention neural networks for time series forecasting," in *International joint conference on neural networks (IJCNN)*, 2021.

[14] Z. Gong, Y. Tang, and J. Liang, "Patchmixer: A patch-mixing architecture for long-term time series forecasting," in *Proceedings of the International Conference on Learning Representations (ICLR 2024)*, vol. 12, 2024.

[15] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proceedings of the AAAI conference on artificial intelligence (AAAI 2021)*, vol. 35, 2021.

[16] H. Wu, J. Xu, J. Wang, and M. Long, "Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting," in *Proceedings of the Conference on Neural Information Processing Systems (NeurIPS 2021)*, vol. 35, 2021.

[17] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?" in *Proceedings of the AAAI conference on artificial intelligence (AAAI 2023)*, vol. 37, 2023.

[18] "Study on enhancement for data collection for nr and endc (release 18)," 2024, standard 3GPP TR 38.843.

Authorized licensed use limited to: International Institute of Information Technology-Raipur. Downloaded on November 30,2025 at 13:55:09 UTC from IEEE Xplore. Restrictions apply.